

Exercise 1: Setting Up JUnit

Calculator.java:

```
package com.exercise;

public class Calculator {
    public int add(int a, int b) {
        return a + b;
    }
}
```

CalculatorTest.java:

```
package com.exercise;

import static org.junit.Assert.assertEquals;
import org.junit.Test;

public class CalculatorTest {
    @Test
    public void testAdd() {
        Calculator calc = new Calculator();
        assertEquals(5, calc.add(2, 3));
    }
}
```

Output:

```
● adityahargunani@Adityas-MacBook demo % mvn test
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:demo >-----
[INFO] Building demo 1.4
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- resources:3.0.2:resources (default-resources) @ demo ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] skip non existing resourceDirectory /Users/adityahargunani/Downloads/JUnit/demo/src/main/resources
[INFO]
[INFO] --- compiler:3.8.0:compile (default-compile) @ demo ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- resources:3.0.2:testResources (default-testResources) @ demo ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] skip non existing resourceDirectory /Users/adityahargunani/Downloads/JUnit/demo/src/test/resources
[INFO]
[INFO] --- compiler:3.8.0:testCompile (default-testCompile) @ demo ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- surefire:2.22.1:test (default-test) @ demo ---
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.CalculatorTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.033 s - in com.example.CalculatorTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 1.303 s
[INFO] Finished at: 2026-06-23T15:19:02+05:30
[INFO] -----
○ adityahargunani@Adityas-MacBook demo %
```